

Wanjun Ning

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🎓 Education

University of Texas	2023.08 – 2027.05
Statistics and Data Science Track Ph.D. Department of Mathematics	Arlington, USA
• Principal supervisor: Prof. Wang, Li • Co-supervisor: Prof. Li, Rencang	
Hong Kong Baptist University	2021.01 – 2023.06
Probability and Mathematical Statistics	
Master of Philosophy, Faculty of Science and Technology	Hong Kong
• Principal supervisor: Prof. Tong, Tiejun • Co-supervisor: Prof. Wu, Jingjin • Graduation Thesis: Analysis and Optimization of User Association in Wireless Networks	
Hong Kong Baptist University	2016.09 – 2020.06
Financial Mathematics	
Bachelor of Science (Honours), Faculty of Science and Technology	Hong Kong

📖 Research Interests

My research focuses on building scalable and efficient machine learning systems for high-dimensional, multimodal data under real-world constraints. I develop methods that integrate computational efficiency, robustness, and interpretability—including graph-regularized sparse learning, multimodal fusion across time series, spectra, text, and images, and attention-based neural networks for feature selection. Ultimately, I aim to design deployable, structure-aware models that advance domain applications and generalize to broader AI systems demanding resource-efficient, trustworthy, and adaptive learning.

Key Words: Multi-Modal Deep Learning; Optimization for ML; Applied Data Science (recommender systems with multimodal fusion, search/ranking feature selection, low-latency embeddings)

📚 Publication

W. Ning, F. Saeed, H. Tang, H. Liu, and L. Wang. “Bridging Neural and Hemodynamic Pathways through EEG-bbNIRS Fusion for Early Alzheimer’s Disease Classification.” Under review, 2026.

Z. Xu, W. Tian, Y. Liu, W. Ning and J. Wu, “A Ring Topology-Based Communication-Efficient Scheme for D2D Wireless Federated Learning,” *In 2023 IEEE Global Communications Conference*, 2023, pp. 2820-2825.

W. Ning, Z. Xu, J. Wu and T. Tong, “Sequence Q-Learning Algorithm for Optimal Mobility-Aware User Association,” *In 2022 IEEE International Conference on Communications (ICC)*, 2022, pp. 726-732.

⚙️ Skills

- Python, MATLAB, C++, R, EViews
- LaTeX, Markdown, Microsoft Word, Excel, PowerPoint

📖 Research Experiences

Research Assistant	2024.02 – now
University of Texas	Arlington, USA
• Adaptive Multimodal Fusion Network for Hybrid EEG-bbNIRS Alzheimer’s Screening – Built a multimodal fusion framework integrating EEG and bbNIRS signals for early Alzheimer’s screening. – Stabilized training with targeted augmentations, adaptive regularization, and learning-rate scheduling. – Analyzed modality contributions, highlighting complementary temporal and metabolic information. – Performance: Achieved 7-12% improvement in AUC over unimodal baselines with robust cross-validation.	

- **Graph-Regularized Sparse Principal Component Analysis**
 - Proposed a unified framework that performs unsupervised dimensionality reduction, interpretable feature selection, and adaptive graph structure learning.
 - Formulated a joint optimization with $\ell_{2,1}$ -norm sparsity and data-driven graph similarity; especially introduced graph ensembles to capture heterogeneous structures.
 - Designed efficient alternating minimization with closed-form updates and convergence guarantees.
 - **Performance:** Produced interpretable feature subsets with 3x faster runtime and stronger clustering or classification accuracy on high-dimensional datasets compared to existing methods.
- **Unrolled Attention-Based Neural Network for Efficient Unsupervised Feature Selection**
 - Modified attention-based neural networks with unrolling and new regularization strategies to improve feature selection in high-dimensional settings.
 - Integrated feature selection with dimensionality reduction across imaging and tabular biomarkers, validated by clinical experts for plausibility.
 - Improved stability and calibration via task-specific objectives and hyperparameter tuning.
 - **Performance:** Achieved 2.5x faster convergence and reduced computation cost compared to baseline feature selection methods, while preserving interpretability.

Research Assistant

2020.06 – 2020.12

Beijing Normal-Hong Kong Baptist University

Zhuhai, China

- Investigated energy efficiency and Quality of Service (QoS) trade-offs in cellular networks by modeling Base Stations (BSs) as processor-sharing queues with vacations.
- Developed and analyzed three BS sleep schemes: isolated, cooperative, and hybrid.
- Proposed a scalable analytical method to evaluate QoS metrics and power consumption.
- Validated results through simulations and conducted extensive experiments to compare scheme performance under different network conditions.
- Skills: MATLAB, Queueing Theory, Network Optimization, Power Efficiency, Simulation.

Business Analysis Intern

2019.06 – 2019.08

China Capital Management Company

Beijing, China

- Analyzed a biotech’ s Phase III oncology asset, synthesizing trial outcomes, market potential, competitive landscape, and strategic implications to inform go-to-market and investment decisions.
- Assessed drug efficacy and safety through clinical data, regulatory submissions, and scientific literature.
- Modeled market share and revenue projections based on disease prevalence and competitor analysis.
- Assisted project manager in presenting key findings and strategic recommendations to senior management.
- Skills: R Programming, Data Analysis, Market Research, Financial Modeling, Pharmaceutical Industry

Awards

Student Travel Award, NSF-CBMS 2026 (Ypsilanti, MI)	2026.05
Student Travel Award, CBMS AMML 2025 (Houston, TX)	2025.12
Dr. Chien-Pai Han Memorial Scholarship	2025
Benny M. McCarley Scholarship	2024
Graduate Research Assistantship/Graduate Teaching Assistantship	2023.08 – now
Graduate Full Scholarship	2021.01 – 2022.12
University Merit Scholarship	2020
The American Mathematical Contest in Modelling (Honourable Mention)	2018

Professional Activity

Fusing EEG and bbNIRS for Alzheimer’s disease classification via data augmentation	
Poster Presentation —AI for Biology & Medicine Symposium, University of North Texas	2025.10
Poster Presentation —CBMS AMML, University of Houston	2025.12
Recent Advances at the Interface of Applied Mathematics and Machine Learning	
Invited Talk —Department of Mathematics, Hong Kong Baptist University	2025.06
Graph-Regularized Sparse Projections for Unsupervised DR and Adaptive Graph Learning	
Poster Presentation —Discover 2025 Student Research Symposium	2025.04
Gene Classification with an OCCA-based Unsupervised Feature Selection Framework	
Invited Paper Presentation —2024 International Chinese Statistical Association (ICSA)	2024.06